

Response time analyses

This vignette contains a full breakdown of the analyses of participants' total response times on the reasoning problems. The most interesting results are synthetically presented in the main manuscript (preprint here). The present document aims to provide a full account of the steps taken to reach these results, and to provide additional analyses that were not included in the manuscript for brevity.

```
library(aphantasiaReasoningViie)
#> Welcome to aphantasiaReasoningViie.
```

Data preparation

First, let's get the clean, analysis-ready data, create cognitive style clusters using OSIVQ scores (see vignette("preparing_data") and vignette("osivq_clusters") for details), and add this classification to the dataset containing response times (RT) data.

```
df_survey <- get_clean_data()$df_survey
# Clustering OSIVQ data
clustering <- cluster_osivq(df_survey)
# Adding named clusters to the survey data
df_survey <- add_named_clusters(
  df_survey, clustering,
  names = c("Spatialiser", "Visualiser", "Verbaliser"),
  levels = c("Visualiser", "Spatialiser", "Verbaliser"),
  contrasts = c("_visualiser", "_spatialiser", "_verbaliser"),
  base = 1
)
# Merging with experiment data
df_expe <-
  dplyr::left_join(
    get_clean_data()$df_expe,
    df_survey |> dplyr::select(id, cluster),
    by = dplyr::join_by("id")
  ) |>
  dplyr::relocate(cluster, .after = "group")
```

For RT analyses, we removed incorrect trials and trials with suspicious RT patterns. These filtering steps were gathered in the `filter_trials_on_rt()` function, which also provides a short summary of the process.

```
df_rt <-
  filter_trials_on_rt(df_expe, verbose = TRUE) |>
  dplyr::select(id, group:strategy_group, problem, category, rt_total)
#>
#> Outlier trials filtration summary
#> 2808 trials before filtering
#> 587 incorrect trials removed (20.9%)
#> 317 trials filtered based on mean + 2 * SD
#> (14.27% of remaining trials)
#> 1904 trials remaining after filtering
```

```

dplyr::glimpse(df_rt)
#> Rows: 1,904
#> Columns: 9
#> $ id                <fct> acdn247721443631359lzb, acdn247721443631359lzb, acdn2~
#> $ group              <fct> Typical, Typical, Typical, Typical, Typical, Typical, T~
#> $ cluster            <fct> Visualiser, Visualiser, Visualiser, Visualiser, Visuali~
#> $ group_2            <fct> Typical, Typical, Typical, Typical, Typical, Typical, T~
#> $ group_3            <fct> Typical, Typical, Typical, Typical, Typical, Typical, T~
#> $ strategy_group     <fct> No visual strategy, No visual strategy, No visual strat~
#> $ problem            <int> 25, 1, 6, 9, 8, 5, 14, 17, 21, 12, 24, 3, 7, 22, 16, 4,~
#> $ category           <fct> Control, Visual, Visual, Visual, Visual, Visual, Spatia~
#> $ rt_total           <dbl> 18.924, 21.771, 21.817, 20.505, 27.169, 22.155, 26.500,~

```

Method

As described in the manuscript:

We fitted GLMMs with Gamma distributions and identity links to account for the skewed distributions of RTs, using the glmmTMB package (McGillcuddy et al., 2025). The models included a grouping variable (VVIQ groups, OSIVQ clusters), Category (visual, spatial, or control) along with their two-way interactions as fixed categorical predictors. Varying slopes and intercepts (“random effects”) have been added for each participant by category and for each problem by grouping variable.

Let’s break this down.

The grouping variables

We used several grouping variables to classify participants, all of which are in the `df_rt` data frame:

- **group** is the 4-group VVIQ classification with an “aphantasia” group (VVIQ = 16), “hypophantasia” group (VVIQ ∈ [17, 32]), “typical” group (VVIQ ∈ [33, 74]), and “hyperphantasia” group (VVIQ ∈ [75, 80]). It was not used in the analyses because the hyperphantasia group was too small (N = 4).
- **group_3** is the same as **group**, but with the hyperphantasia group merged with the typical group. It is referred to as “VVIQ 3 groups” in the manuscript results.
- **group_2** is the binary classification often used in aphantasia literature that additionally merges the hypophantasia group with the aphantasia group. It is referred to as “VVIQ 2 groups” in the manuscript results.
- **cluster** is the cognitive style classification obtained from clustering OSIVQ scores. It contains three groups: “Visualiser”, “Spatialiser”, and “Verbaliser”. It is referred to as “OSIVQ 3 clusters” in the manuscript results.
- **strategy_group** is a classification based on the self-reported strategies used by the participants to solve the problems. It focuses on whether participants used a visual mental imagery strategy and contains two groups: “Visual strategy user” and “No visual strategy”. It is not reported in the manuscript because it was suggested to us after submission by colleagues at a conference.

The same modelling pipeline was therefore applied four times, once for each of the last four grouping variables.

The modelling pipeline

We created a small `build_formula()` helper function to write the formula easily, as we used the same model structure a lot of times:

```

build_formula("rt_total", "group_2")
#> rt_total ~ group_2 * category + (category | id) + (group_2 |
#>     problem)
#> <environment: 0x00000175f8fa8238>

build_formula("rt_total", "cluster")
#> rt_total ~ cluster * category + (category | id) + (cluster |
#>     problem)
#> <environment: 0x00000175f7127770>

```

The models were fitted using the `glmmTMB::glmmTMB()` function (and a little `set_ranef_prior()` helper function to set a weakly informative prior on the random effects, which helps with convergence and singularity issues). After model fit, we checked for potential singularity issues and model performance using the `get_singularity()` and `get_performance()` functions created for the occasion (which are convenient wrappers around the *performance* package).

Finally, we tested our hypotheses with marginal contrasts. As stated in the manuscript:

Due to the way that variance is partitioned in GLMMs (Rights & Sterba, 2019), there does not exist an agreed-upon way to calculate standard effect sizes for individual terms such as main effects or interactions in these models. Thus, in line with general recommendations on how to report effect sizes (e.g., Pek & Flora, 2018), we report and analyse unstandardised effect sizes for post-hoc tests in the form of estimated marginal contrasts (i.e. differences in model-estimated marginal means, hereinafter denoted Δ), in seconds for RTs or as odds ratios for accuracies. To answer our hypotheses, we planned to analyse contrasts between groups, contrasts between categories for each group separately, and interaction contrasts testing the differences in category contrasts between the groups.

This task was performed with the `report_contrast()` function, which is a wrapper around several functions from the *emmeans* package. We used it to compute response time contrasts between groups, contrasts between categories within each group, and interaction contrasts (differences in category contrasts between groups).

Here we go!

Results

VVIQ 2 groups

```

m_rt_vviq_2 <-
  glmmTMB::glmmTMB(
    data      = df_rt,
    formula   = build_formula("rt_total", "group_2"),
    family    = Gamma(link = "identity"),
    prior     = set_ranef_prior(70)
  )

# Singularity
m_rt_vviq_2 |> get_singularity()
#> The model is not singular, parameter estimates are trustworthy.

# Performance
m_rt_vviq_2 |> get_performance() |> knitr::kable(align = "c")

```

AIC	BIC	R2 (cond.)	R2 (marg.)	ICC	RMSE
12508.272	12597.100	0.998	0.027	0.998	5.910

```
# Group contrasts
m_rt_vviiq_2 |> report_contrast(~ group_2) |> knitr::kable()
#> NOTE: Results may be misleading due to involvement in interactions
```

Contrast	Difference	95% CI	p.value
Aphantasia - Typical	-0.628	[-2.96, 1.7]	0.598

```
# Category contrasts within groups
m_rt_vviiq_2 |> report_contrast(~ category | group_2) |> knitr::kable()
```

Contrast	group_2	Difference	95% CI	p.value
Control - Spatial	Aphantasia	-0.066	[-2.22, 2.09]	0.997
Control - Visual	Aphantasia	-1.685	[-3.82, 0.45]	0.153
Spatial - Visual	Aphantasia	-1.618	[-3.97, 0.73]	0.241
Control - Spatial	Typical	0.037	[-1.83, 1.91]	0.999
Control - Visual	Typical	-2.411	[-4.27, -0.55]	0.007
Spatial - Visual	Typical	-2.448	[-4.52, -0.38]	0.015

```
# Interaction contrasts
m_rt_vviiq_2 |> report_contrast(~ category * group_2, interaction = TRUE) |>
  knitr::kable()
```

Category contrast	group_2_pairwise	Difference	95% CI	p.value
Control - Spatial	Aphantasia - Typical	-0.103	[-1.51, 1.3]	0.885
Control - Visual	Aphantasia - Typical	0.726	[-0.63, 2.08]	0.294
Spatial - Visual	Aphantasia - Typical	0.829	[-0.94, 2.6]	0.358

VVIQ 3 groups

```
m_rt_vviiq_3 <-
  glmmTMB::glmmTMB(
    data = df_rt,
    formula = build_formula("rt_total", "group_3"),
    family = Gamma(link = "identity"),
    prior = set_ranef_prior(70)
  )

m_rt_vviiq_3 |> get_singularity()
#> The model is not singular, parameter estimates are trustworthy.

m_rt_vviiq_3 |> get_performance() |> knitr::kable(align = "c")
```

AIC	BIC	R2 (cond.)	R2 (marg.)	ICC	RMSE
12552.989	12675.127	0.998	0.054	0.998	5.909

```
m_rt_vviiq_3 |> report_contrast(~ group_3) |> knitr::kable()
#> NOTE: Results may be misleading due to involvement in interactions
```

Contrast	Difference	95% CI	p.value
Aphantasia - Hypophantasia	3.160	[-1.06, 7.38]	0.185
Aphantasia - Typical	0.555	[-2.6, 3.71]	0.911
Hypophantasia - Typical	-2.605	[-6.45, 1.24]	0.251

```
m_rt_vviiq_3 |> report_contrast(~ category | group_3) |> knitr::kable()
```

Contrast	group_3	Difference	95% CI	p.value
Control - Spatial	Aphantasia	-0.061	[-2.62, 2.5]	0.998
Control - Visual	Aphantasia	-1.345	[-3.85, 1.16]	0.420
Spatial - Visual	Aphantasia	-1.284	[-4.09, 1.52]	0.532
Control - Spatial	Hypophantasia	-0.066	[-2.83, 2.7]	0.998
Control - Visual	Hypophantasia	-2.297	[-5, 0.41]	0.115
Spatial - Visual	Hypophantasia	-2.231	[-5.4, 0.94]	0.225
Control - Spatial	Typical	0.037	[-1.82, 1.9]	0.999
Control - Visual	Typical	-2.434	[-4.28, -0.59]	0.006
Spatial - Visual	Typical	-2.470	[-4.52, -0.42]	0.013

```
m_rt_vviiq_3 |> report_contrast(~ category * group_3, interaction = TRUE) |>
knitr::kable()
```

Category contrast	group_3_pairwise	Difference	95% CI	p.value
Control - Spatial	Aphantasia - Hypophantasia	0.006	[-2.11, 2.12]	0.996
Control - Visual	Aphantasia - Hypophantasia	0.953	[-1.05, 2.95]	0.351
Spatial - Visual	Aphantasia - Hypophantasia	0.947	[-1.71, 3.6]	0.484
Control - Spatial	Aphantasia - Typical	-0.097	[-1.79, 1.59]	0.910
Control - Visual	Aphantasia - Typical	1.089	[-0.53, 2.71]	0.187
Spatial - Visual	Aphantasia - Typical	1.186	[-0.9, 3.27]	0.264
Control - Spatial	Hypophantasia - Typical	-0.103	[-2.05, 1.85]	0.917
Control - Visual	Hypophantasia - Typical	0.136	[-1.73, 2]	0.886
Spatial - Visual	Hypophantasia - Typical	0.239	[-2.21, 2.69]	0.848

OSIVQ 3 clusters

```
m_rt_osivq <-
glmmTMB::glmmTMB(
  data = df_rt,
  formula = build_formula("rt_total", "cluster"),
  family = Gamma(link = "identity"),
  prior = set_ranef_prior(70)
)
```

```
m_rt_osivq |> get_singularity()
#> The model is not singular, parameter estimates are trustworthy.

m_rt_osivq |> get_performance() |> knitr::kable(align = "c")
```

AIC	BIC	R2 (cond.)	R2 (marg.)	ICC	RMSE
12555.203	12677.341	0.998	0.032	0.998	5.907

```
m_rt_osivq |> report_contrast(~ cluster) |> knitr::kable()
#> NOTE: Results may be misleading due to involvement in interactions
```

Contrast	Difference	95% CI	p.value
Visualiser - Spatialiser	1.265	[-2.62, 5.15]	0.726
Visualiser - Verbaliser	0.667	[-2.4, 3.74]	0.867
Spatialiser - Verbaliser	-0.597	[-4.49, 3.3]	0.931

```
m_rt_osivq |> report_contrast(~ category | cluster) |> knitr::kable()
```

Contrast	Cluster	Difference	95% CI	p.value
Control - Spatial	Visualiser	-0.266	[-2.2, 1.67]	0.944
Control - Visual	Visualiser	-2.890	[-4.8, -0.98]	0.001
Spatial - Visual	Visualiser	-2.624	[-4.8, -0.44]	0.013
Control - Spatial	Spatialiser	0.206	[-2.43, 2.84]	0.982
Control - Visual	Spatialiser	-1.365	[-3.94, 1.21]	0.429
Spatial - Visual	Spatialiser	-1.572	[-4.58, 1.44]	0.440
Control - Spatial	Verbaliser	0.167	[-2.08, 2.41]	0.983
Control - Visual	Verbaliser	-1.540	[-3.75, 0.67]	0.231
Spatial - Visual	Verbaliser	-1.707	[-4.16, 0.74]	0.231

```
m_rt_osivq |> report_contrast(~ category * cluster, interaction = TRUE) |>
knitr::kable()
```

Category contrast	Cluster contrast	Difference	95% CI	p.value
Control - Spatial	Visualiser - Spatialiser	-0.473	[-2.42, 1.47]	0.634
Control - Visual	Visualiser - Spatialiser	-1.525	[-3.38, 0.33]	0.107
Spatial - Visual	Visualiser - Spatialiser	-1.052	[-3.5, 1.39]	0.399
Control - Spatial	Visualiser - Verbaliser	-0.433	[-2.04, 1.18]	0.598
Control - Visual	Visualiser - Verbaliser	-1.351	[-2.9, 0.2]	0.087
Spatial - Visual	Visualiser - Verbaliser	-0.917	[-2.91, 1.07]	0.366
Control - Spatial	Spatialiser - Verbaliser	0.039	[-1.88, 1.95]	0.968
Control - Visual	Spatialiser - Verbaliser	0.174	[-1.62, 1.97]	0.849
Spatial - Visual	Spatialiser - Verbaliser	0.135	[-2.28, 2.54]	0.913

Strategy groups

```
m_rt_strat <-
glmmTMB::glmmTMB(
```

```

data      = df_rt,
formula   = build_formula("rt_total", "strategy_group"),
family    = Gamma(link = "identity"),
prior     = set_ranef_prior(70)
)

m_rt_strat |> get_singularity()
#> The model is not singular, parameter estimates are trustworthy.

m_rt_strat |> get_performance() |> knitr::kable(align = "c")

```

AIC	BIC	R2 (cond.)	R2 (marg.)	ICC	RMSE
12505.623	12594.450	0.998	0.031	0.998	5.904

```

m_rt_strat |> report_contrast(~ strategy_group) |> knitr::kable()
#> NOTE: Results may be misleading due to involvement in interactions

```

Contrast	Difference	95% CI	p.value
Visual strategy user - No visual strategy	-0.895	[-3.21, 1.42]	0.449

```

m_rt_strat |> report_contrast(~ category | strategy_group) |> knitr::kable()

```

Contrast	strategy_group	Difference	95% CI	p.value
Control - Spatial	Visual Strategy User	-0.032	[-1.88, 1.82]	0.999
Control - Visual	Visual Strategy User	-2.523	[-4.36, -0.68]	0.004
Spatial - Visual	Visual Strategy User	-2.491	[-4.55, -0.43]	0.013
Control - Spatial	No Visual Strategy	0.018	[-2.19, 2.23]	1.000
Control - Visual	No Visual Strategy	-1.605	[-3.8, 0.59]	0.199
Spatial - Visual	No Visual Strategy	-1.623	[-4.01, 0.77]	0.249

```

m_rt_strat |> report_contrast(~ category * strategy_group, interaction = TRUE) |>
  knitr::kable()

```

Category contrast	strategy_group_pairwise	Difference	95% CI	p.value
Control - Spatial	Visual Strategy User - No Visual Strategy	-0.050	[-1.47, 1.37]	0.945
Control - Visual	Visual Strategy User - No Visual Strategy	-0.918	[-2.29, 0.46]	0.191
Spatial - Visual	Visual Strategy User - No Visual Strategy	-0.868	[-2.65, 0.91]	0.340

Visualisation

The figures showing the distribution of response times displayed in the manuscript were created with the `plot_superb_raincloud()` function from the package, which wraps functions from the *ggplot2* and *superb* packages to create nice visualisations easily. A little `add_significance()` helper function was also created to add significance stars to the plots.

```

library(patchwork)
library(superb)

```

```

pr1 <-
  plot_superb_raincloud(
    df_rt, rt_total, group_2,
    title = "VVIQ 2 groups", y_title = "Mean total RT (s)",
    plot.background = ggplot2::element_rect(fill = "white", colour = NA)
  ) +
  add_significance(
    size_star = 4,
    tibble::tibble(
      x_star = 2,
      y_star = 30,
      stars = "**",
      x_line = .data$x_star - 0.14,
      x_line_end = .data$x_star + 0.14,
      y_line = 29.5
    )
  ) +
  add_significance(
    size_star = 4,
    tibble::tibble(
      x_star = 2.07,
      y_star = 28,
      stars = "*",
      x_line = .data$x_star - 0.07,
      x_line_end = .data$x_star + 0.07,
      y_line = 27.5
    )
  )
)

pr2 <-
  plot_superb_raincloud(
    df_rt, rt_total, group_3,
    title = "VVIQ 3 groups", y_title = "Mean total RT (s)",
    plot.background = ggplot2::element_rect(fill = "white", colour = NA)
  ) +
  add_significance(
    size_star = 4,
    tibble::tibble(
      x_star = 3,
      y_star = 30,
      stars = "**",
      x_line = .data$x_star - 0.14,
      x_line_end = .data$x_star + 0.14,
      y_line = 29.5
    )
  ) +
  add_significance(
    size_star = 4,
    tibble::tibble(
      x_star = 3.07,
      y_star = 28,
      stars = "*",
      x_line = .data$x_star - 0.07,

```



```

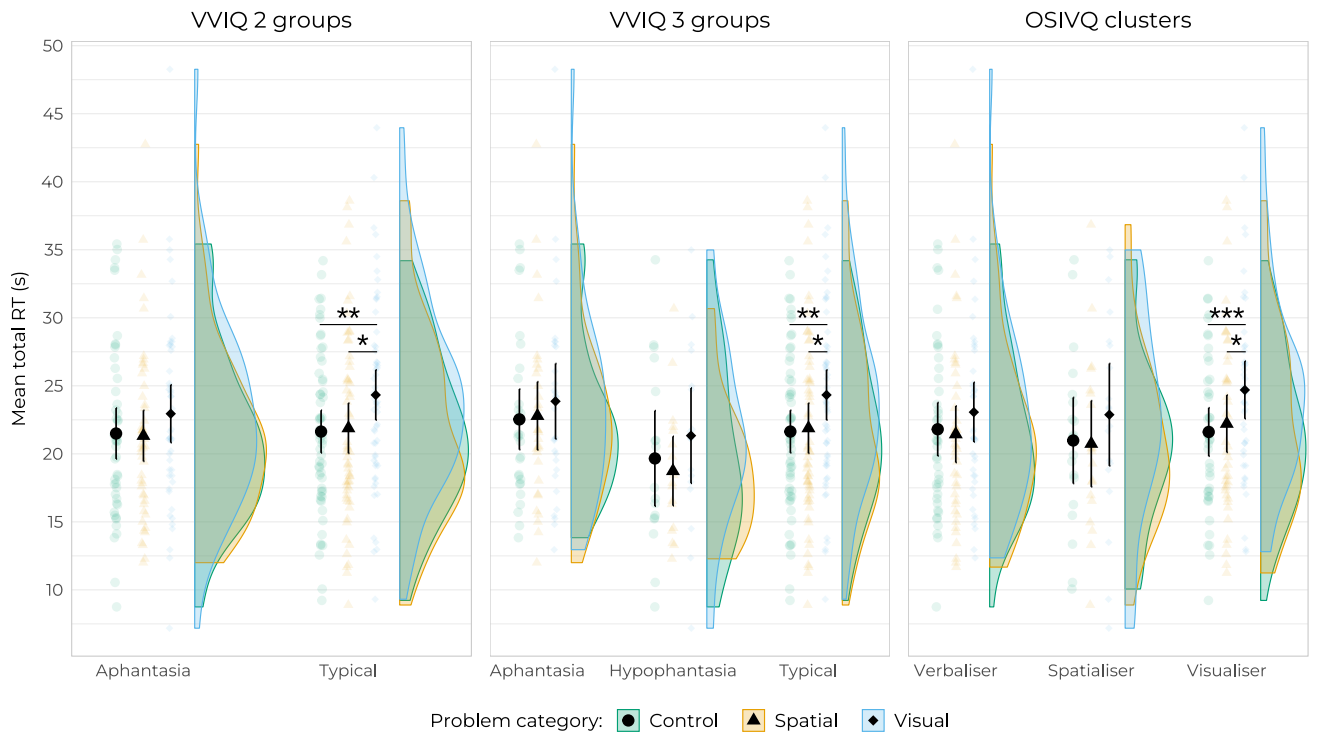
    x_line_end = .data$x_star + 0.07,
    y_line = 27.5
  )
)

pr3 <-
  plot_superb_raincloud(
    df_rt, rt_total, cluster,
    title = "OSIVQ clusters", y_title = "Mean total RT (s)",
    plot.background = ggplot2::element_rect(fill = "white", colour = NA)
  ) +
  add_significance(
    size_star = 4,
    tibble::tibble(
      x_star = 3,
      y_star = 30,
      stars = "***",
      x_line = .data$x_star - 0.14,
      x_line_end = .data$x_star + 0.14,
      y_line = 29.5
    )
  ) +
  add_significance(
    size_star = 4,
    tibble::tibble(
      x_star = 3.07,
      y_star = 28,
      stars = "*",
      x_line = .data$x_star - 0.07,
      x_line_end = .data$x_star + 0.07,
      y_line = 27.5
    )
  )

pr <- pr1 + pr2 + pr3 +
  patchwork::plot_layout(axes = "collect", guides = "collect") &
  ggplot2::theme(legend.position = "bottom")

plot(pr)

```



All done!

```

#> - Session info -----
#> setting value
#> version R version 4.5.0 (2025-04-11 ucrt)
#> os      Windows 11 x64 (build 26100)
#> system  x86_64, mingw32
#> ui      RTerm
#> language (EN)
#> collate French_France.utf8
#> ctype   French_France.utf8
#> tz      Europe/Paris
#> date    2025-09-17
#> pandoc  3.6.3 @ C:/Program Files/RStudio/resources/app/bin/quarto/bin/tools/ (via rmarkdown)
#>
#> - Packages -----
#> ! package          * version      date (UTC) lib source
#> P abind             1.4-8        2024-09-12 [?] CRAN (R 4.5.0)
#> P aphantasiaReasoningViie * 0.0.0.9000 2025-09-16 [1] local
#> P assertthat        0.2.1        2019-03-21 [?] CRAN (R 4.5.0)
#> P boot              1.3-31       2024-08-28 [3] CRAN (R 4.5.0)
#> P class             7.3-23       2025-01-01 [3] CRAN (R 4.5.0)
#> P cli               3.6.5        2025-04-23 [?] CRAN (R 4.5.0)
#> P clue              0.3-66       2024-11-13 [?] CRAN (R 4.5.0)
#> P cluster           2.1.8.1      2025-03-12 [3] CRAN (R 4.5.0)
#> P clusterCrit       1.3.0        2023-11-23 [?] CRAN (R 4.5.0)
#> P clValid           0.7          2021-02-14 [?] CRAN (R 4.5.0)
#> P coda              0.19-4.1     2024-01-31 [?] CRAN (R 4.5.0)
#> P combinat          0.0-8        2012-10-29 [?] CRAN (R 4.5.0)
#> P crayon            1.5.3        2024-06-20 [?] CRAN (R 4.5.0)
#> P curl              6.3.0        2025-06-06 [?] RSPM
#> P diceR             3.1.0        2025-06-19 [?] RSPM (R 4.5.0)
#> P digest            0.6.37       2024-08-19 [?] CRAN (R 4.5.0)
#> P dplyr             1.1.4        2023-11-17 [?] CRAN (R 4.5.0)
#> P e1071             1.7-16       2024-09-16 [?] CRAN (R 4.5.0)
#> P emmeans           1.11.1       2025-05-04 [?] CRAN (R 4.5.0)
#> P estimability      1.5.1        2024-05-12 [?] CRAN (R 4.5.0)
#> P evaluate          1.0.4        2025-06-18 [?] RSPM (R 4.5.0)
#> P farver            2.1.2        2024-05-13 [?] CRAN (R 4.5.0)
#> P fastmap           1.2.0        2024-05-15 [?] CRAN (R 4.5.0)
#> P forcats           1.0.0        2023-01-29 [?] CRAN (R 4.5.0)
#> P foreign           0.8-90       2025-03-31 [3] CRAN (R 4.5.0)
#> P generics          0.1.4        2025-05-09 [?] CRAN (R 4.5.0)
#> P ggplot2           3.5.2        2025-04-09 [?] CRAN (R 4.5.0)
#> P glmmTMB           1.1.11       2025-04-02 [?] CRAN (R 4.5.0)
#> P glue              1.8.0        2024-09-30 [?] CRAN (R 4.5.0)
#> P gtable            0.3.6        2024-10-25 [?] CRAN (R 4.5.0)
#> P haven             2.5.5        2025-05-30 [?] RSPM (R 4.5.0)
#> P highr             0.11         2024-05-26 [?] CRAN (R 4.5.0)
#> P hms               1.1.3        2023-03-21 [?] CRAN (R 4.5.0)
#> P htmltools         0.5.8.1      2024-04-04 [?] CRAN (R 4.5.0)
#> P httpuv            1.6.16       2025-04-16 [?] CRAN (R 4.5.0)
#> P insight           1.3.1        2025-06-30 [?] RSPM (R 4.5.0)
#> P jsonlite          2.0.0        2025-03-27 [?] CRAN (R 4.5.0)
#> P klaR              1.7-3        2023-12-13 [?] CRAN (R 4.5.0)
#> P knitr             1.50         2025-03-16 [?] CRAN (R 4.5.0)

```

#> P labelled	2.14.1	2025-05-06	[?]	CRAN	(R 4.5.0)
#> P later	1.4.2	2025-04-08	[?]	CRAN	(R 4.5.0)
#> P lattice	0.22-7	2025-04-02	[?]	CRAN	(R 4.5.0)
#> P lifecycle	1.0.4	2023-11-07	[?]	CRAN	(R 4.5.0)
#> P lme4	1.1-37	2025-03-26	[?]	CRAN	(R 4.5.0)
#> P lsr	0.5.2	2021-12-01	[1]	RSPM	(R 4.5.0)
#> P magrittr	2.0.3	2022-03-30	[?]	CRAN	(R 4.5.0)
#> P MASS	7.3-65	2025-02-28	[3]	CRAN	(R 4.5.0)
#> P Matrix	1.7-3	2025-03-11	[3]	CRAN	(R 4.5.0)
#> P mclust	6.1.1	2024-04-29	[?]	CRAN	(R 4.5.0)
#> P mgcv	1.9-3	2025-04-04	[?]	CRAN	(R 4.5.0)
#> P mime	0.13	2025-03-17	[?]	CRAN	(R 4.5.0)
#> P miniUI	0.1.2	2025-04-17	[?]	CRAN	(R 4.5.0)
#> P minqa	1.2.8	2024-08-17	[?]	CRAN	(R 4.5.0)
#> P mvtnorm	1.3-3	2025-01-10	[?]	CRAN	(R 4.5.0)
#> P nlme	3.1-168	2025-03-31	[3]	CRAN	(R 4.5.0)
#> P nloptr	2.2.1	2025-03-17	[?]	CRAN	(R 4.5.0)
#> P numDeriv	2016.8-1.1	2019-06-06	[?]	CRAN	(R 4.5.0)
#> P patchwork	* 1.3.1	2025-06-21	[?]	RSPM	(R 4.5.0)
#> P performance	0.14.0	2025-05-22	[?]	CRAN	(R 4.5.0)
#> P pillar	1.10.2	2025-04-05	[?]	CRAN	(R 4.5.0)
#> P pkgconfig	2.0.3	2019-09-22	[?]	CRAN	(R 4.5.0)
#> P plyr	1.8.9	2023-10-02	[?]	CRAN	(R 4.5.0)
#> P promises	1.3.3	2025-05-29	[?]	RSPM	
#> P proxy	0.4-27	2022-06-09	[?]	CRAN	(R 4.5.0)
#> P purrr	1.0.4	2025-02-05	[?]	CRAN	(R 4.5.0)
#> P questionr	0.8.1	2025-06-10	[?]	RSPM	(R 4.5.0)
#> P R6	2.6.1	2025-02-15	[?]	CRAN	(R 4.5.0)
#> P rbibutils	2.3	2024-10-04	[?]	CRAN	(R 4.5.0)
#> P RColorBrewer	1.1-3	2022-04-03	[?]	CRAN	(R 4.5.0)
#> P Rcpp	1.0.14	2025-01-12	[?]	CRAN	(R 4.5.0)
#> P Rdpack	2.6.4	2025-04-09	[?]	CRAN	(R 4.5.0)
#> P reformulas	0.4.1	2025-04-30	[?]	CRAN	(R 4.5.0)
#> P reshape2	1.4.4	2020-04-09	[?]	CRAN	(R 4.5.0)
#> P rlang	1.1.6	2025-04-11	[?]	CRAN	(R 4.5.0)
#> P rmarkdown	2.29	2024-11-04	[?]	CRAN	(R 4.5.0)
#> P rrappl	1.2.7	2024-06-26	[1]	RSPM	(R 4.5.0)
#> P rstudioapi	0.17.1	2024-10-22	[?]	CRAN	(R 4.5.0)
#> P scales	1.4.0	2025-04-24	[?]	CRAN	(R 4.5.0)
#> P sessioninfo	1.2.2	2021-12-06	[?]	RSPM	(R 4.5.0)
#> P shiny	1.10.0	2024-12-14	[?]	CRAN	(R 4.5.0)
#> P shinyBS	0.61.1	2022-04-17	[1]	RSPM	(R 4.5.0)
#> P showtext	0.9-7	2024-03-02	[?]	CRAN	(R 4.5.0)
#> P showtextdb	3.0	2020-06-04	[?]	CRAN	(R 4.5.0)
#> P stringi	1.8.7	2025-03-27	[?]	CRAN	(R 4.5.0)
#> P stringr	1.5.1	2023-11-14	[?]	CRAN	(R 4.5.0)
#> P superb	* 1.0.0	2025-08-18	[1]	RSPM	(R 4.5.0)
#> P sysfonts	0.8.9	2024-03-02	[?]	CRAN	(R 4.5.0)
#> P tibble	3.3.0	2025-06-08	[?]	RSPM	
#> P tidyr	1.3.1	2024-01-24	[?]	CRAN	(R 4.5.0)
#> P tidyselect	1.2.1	2024-03-11	[?]	CRAN	(R 4.5.0)
#> PD TMB	1.9.17	2025-03-10	[?]	CRAN	(R 4.5.0)
#> P vctrs	0.6.5	2023-12-01	[?]	CRAN	(R 4.5.0)
#> P withr	3.0.2	2024-10-28	[?]	CRAN	(R 4.5.0)

```

#> P   xfun                0.52      2025-04-02 [?] CRAN (R 4.5.0)
#> P   xtable              1.8-4      2019-04-21 [?] CRAN (R 4.5.0)
#> P   yaml                2.3.10     2024-07-26 [?] CRAN (R 4.5.0)
#>
#> [1] C:/Users/Mael/AppData/Local/R/cache/R/renv/library/aphantasiaReasoningViie-d318fbb0/windows/R-4
#> [2] C:/Users/Mael/AppData/Local/R/cache/R/renv/sandbox/windows/R-4.5/x86_64-w64-mingw32/9bc714e4
#> [3] C:/Program Files/R/R-4.5.0/library
#>
#> P -- Loaded and on-disk path mismatch.
#> D -- DLL MD5 mismatch, broken installation.
#>
#> -----

```